



MINISTRY OF EDUCATION, CULTURE AND RESEARCH

OF THE REPUBLIC OF MOLDOVA

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Report

on laboratory work no. 1

of Operating Systems

Topic: The OS as a Resource Manager

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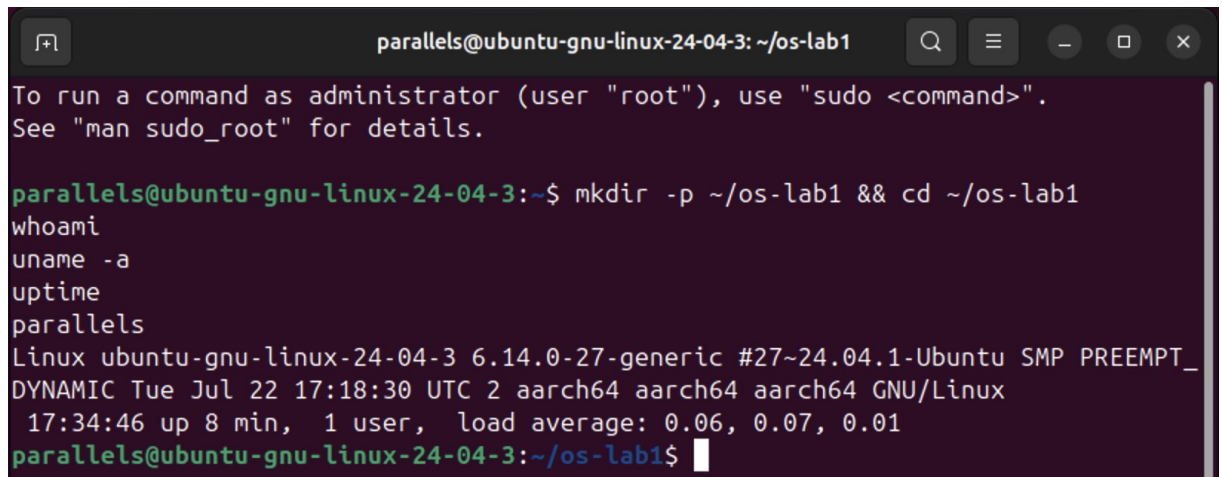
Purpose of the work

The purpose of this laboratory work is to observe the operating system acting as a resource manager on a Linux virtual machine (Ubuntu 24.04.3 ARM64, running in Parallels on Apple Silicon). Using terminal commands, the four core resources managed by any OS were examined: files, processes, memory and devices.

Before you start

```
$ whoami
parallels
$ uname -a
Linux ubuntu-gnu-linux-24-04-3 6.14.0-27-generic #27~24.04.1-Ubuntu SMP
PREEMPT_DYNAMIC Tue Jul 22 17:18:30 UTC 2 aarch64 aarch64 aarch64 GNU/Linux
$ uptime
17:34:46 up 8 min, 1 user, load average: 0.06, 0.07, 0.01
```

The machine runs Linux kernel 6.14 on an ARM64 (aarch64) CPU.

A screenshot of a terminal window titled 'parallels@ubuntu-gnu-linux-24-04-3: ~/os-lab1'. The terminal shows a series of commands and their outputs. First, a message explains how to use 'sudo'. Then, the command 'mkdir -p ~/os-lab1 && cd ~/os-lab1' is executed. Following this, the commands 'whoami', 'uname -a', and 'uptime' are run, producing the same output as shown in the text block above. The prompt at the bottom is 'parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1\$' with a cursor.

```
parallels@ubuntu-gnu-linux-24-04-3: ~/os-lab1
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

parallels@ubuntu-gnu-linux-24-04-3:~$ mkdir -p ~/os-lab1 && cd ~/os-lab1
whoami
uname -a
uptime
parallels
Linux ubuntu-gnu-linux-24-04-3 6.14.0-27-generic #27~24.04.1-Ubuntu SMP PREEMPT_
DYNAMIC Tue Jul 22 17:18:30 UTC 2 aarch64 aarch64 aarch64 GNU/Linux
17:34:46 up 8 min, 1 user, load average: 0.06, 0.07, 0.01
parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1$
```

Figure 1.1 – Creating the working folder and identifying the user, kernel and uptime

Part 1. Files and directories

```
parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1$ pwd
ls -la /
ls -la ~
cd /etc && ls | head
/home/parallels/os-lab1
total 3992676
drwxr-xr-x 23 root root      4096 Aug 18  2025 .
drwxr-xr-x 23 root root      4096 Aug 18  2025 ..
lrwxrwxrwx  1 root root         7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x  2 root root      4096 Feb 26  2024 bin.usr-is-merged
drwxr-xr-x  4 root root      4096 Sep 28 17:26 boot
dr-xr-xr-x  2 root root      4096 Aug  6  2025 cdrom
drwxr-xr-x 19 root root     4180 Sep 28 17:26 dev
drwxr-xr-x 138 root root    12288 Sep 28 17:26 etc
drwxr-xr-x  3 root root      4096 Aug 14  2025 home
lrwxrwxrwx  1 root root         7 Apr 22  2024 lib -> usr/lib
drwxr-xr-x  2 root root      4096 Apr  8  2024 lib.usr-is-merged
drwx----- 2 root root    16384 Aug 14  2025 lost+found
drwxr-xr-x  4 root root      4096 Sep 28 17:26 media
drwxr-xr-x  2 root root      4096 Aug  6  2025 mnt
drwxr-xr-x  2 root root      4096 Aug  6  2025 opt
dr-xr-xr-x 256 root root      0 Sep 28 17:26 proc
drwx----- 5 root root      4096 Aug 18  2025 root
drwxr-xr-x 35 root root      900 Sep 28 17:28 run
lrwxrwxrwx  1 root root         8 Apr 22  2024/sbin -> usr/sbin
drwxr-xr-x  2 root root      4096 Mar 31  2024/sbin.usr-is-merged
drwxr-xr-x 11 root root      4096 Aug  6  2025 snap
drwxr-xr-x  2 root root      4096 Aug  6  2025 srv
-rw-----  1 root root 4088397824 Aug 14  2025 swap.img
dr-xr-xr-x 13 root root      0 Sep 28 17:35 sys
drwxrwxrwt 16 root root      4096 Sep 28 17:36 tmp
drwxr-xr-x 11 root root      4096 Aug  6  2025 usr
drwxr-xr-x 14 root root      4096 Aug 14  2025 var
total 84
drwxr-xr-x 17 parallels parallels 4096 Sep 28 17:35 .
drwxr-xr-x  3 root      root      4096 Aug 14  2025 ..
-rw-----  1 parallels parallels  86 Sep 28 17:35 .bash_history
-rw-r--r--  1 parallels parallels 220 Mar 31  2024 .bash_logout
-rw-r--r--  1 parallels parallels 3771 Mar 31  2024 .bashrc
```

Figure 1.2 – Contents of the root directory (ls -la /)

```

drwxr-xr-x  2 root root      4096 Aug  6  2025 srv
-rw-----  1 root root 4088397824 Aug 14  2025 swap.img
dr-xr-xr-x 13 root root          0 Sep 28 17:35 sys
drwxrwxrwt 16 root root      4096 Sep 28 17:36 tmp
drwxr-xr-x 11 root root      4096 Aug  6  2025 usr
drwxr-xr-x 14 root root      4096 Aug 14  2025 var
total 84
drwxr-x--- 17 parallels parallels 4096 Sep 28 17:35 .
drwxr-xr-x  3 root      root      4096 Aug 14  2025 ..
-rw-----  1 parallels parallels   86 Sep 28 17:35 .bash_history
-rw-r--r--  1 parallels parallels  220 Mar 31  2024 .bash_logout
-rw-r--r--  1 parallels parallels 3771 Mar 31  2024 .bashrc
drwx----- 11 parallels parallels 4096 Aug 26  2025 .cache
drwx----- 12 parallels parallels 4096 Sep 28 17:27 .config
drwxr-xr-x  2 parallels parallels 4096 Sep 28 17:26 Desktop
drwxr-xr-x  2 parallels parallels 4096 Aug 14  2025 Documents
drwxr-xr-x  2 parallels parallels 4096 Aug 14  2025 Downloads
drwx-----  4 parallels parallels 4096 Aug 14  2025 .local
drwxr-xr-x  2 parallels parallels 4096 Aug 14  2025 Music
drwxrwxr-x  2 parallels parallels 4096 Sep 28 17:34 os-lab1
drwxr-xr-x  2 parallels parallels 4096 Sep 28 17:27 .parallels
drwxr-xr-x  2 parallels parallels 4096 Aug 14  2025 Pictures
-rw-r--r--  1 parallels parallels   807 Mar 31  2024 .profile
drwxr-xr-x  2 parallels parallels 4096 Aug 14  2025 Public
drwx-----  3 parallels parallels 4096 Aug 14  2025 snap
drwx-----  2 parallels parallels 4096 Sep 28 17:26 .ssh
drwxr-xr-x  2 parallels parallels 4096 Aug 14  2025 Templates
drwxr-xr-x  2 parallels parallels 4096 Aug 14  2025 Videos
adduser.conf
alsa
alternatives
anacrontab
apg.conf
apm
apparmor
apparmor.d
appport
apt
parallels@ubuntu-gnu-linux-24-04-3:/etc$

```

Figure 1.3 – Home directory listing and the first entries of /etc

```

parallels@ubuntu-gnu-linux-24-04-3:/etc$ cd ~/os-lab1
mkdir demo && cd demo
echo "hello operating systems" > note.txt
cp note.txt copy.txt
mv copy.txt renamed.txt
ls -l
rm renamed.txt
total 8
-rw-rw-r-- 1 parallels parallels 24 Sep 28 17:37 note.txt
-rw-rw-r-- 1 parallels parallels 24 Sep 28 17:37 renamed.txt
parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$

```

Figure 1.4 – Creating, copying, renaming and removing files

```

parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$ ls -l note.txt
chmod 600 note.txt
ls -l note.txt
chmod 644 note.txt
-rw-rw-r-- 1 parallels parallels 24 Sep 28 17:37 note.txt
-rw----- 1 parallels parallels 24 Sep 28 17:37 note.txt
parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$

```

Figure 1.5 – Permissions of note.txt before and after chmod 600

Interesting output:

```
$ ls -l
total 8
-rw-rw-r-- 1 parallels parallels 24 Sep 28 17:37 note.txt
-rw-rw-r-- 1 parallels parallels 24 Sep 28 17:37 renamed.txt

$ chmod 600 note.txt && ls -l note.txt
-rw----- 1 parallels parallels 24 Sep 28 17:37 note.txt
```

(1) Who owns the files, and which group?

The files are owned by user `parallels`, and their group is also `parallels`. On Ubuntu, every user gets a private group with the same name as the user.

(2) What do the ten characters mean after `chmod 600`?

They are `-rw-----`:

- `-` means a regular file (`d` would mean a directory, `l` a symbolic link).
- `rw-` means the owner can read and write, but not execute.
- `---` means the group has no permissions.
- `---` means others have no permissions.

So only `parallels` can read or change `note.txt`. Before the `chmod` it was `-rw-rw-r--`: the group could also write, and others could read.

(3) Two directories directly under `/`:

- `/etc` holds system-wide configuration files, such as `adduser.conf`, `apt/` and `apparmor/`.
- `/home` holds users' personal directories (here, `/home/parallels`).

All top-level entries in `/` are owned by `root`, while everything inside `~` is owned by `parallels`. The OS separates system files from user files by ownership.

Part 2. Processes

```
parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$ ps aux | head
ps aux | wc -l
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.3 23052 13228 ?        Ss   17:26   0:00 /sbin/init splash
root         2  0.0  0.0      0     0 ?        S    17:26   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        S    17:26   0:00 [pool_workqueue_release]
root         4  0.0  0.0      0     0 ?        I<   17:26   0:00 [kworker/R-rcu_gp]
root         5  0.0  0.0      0     0 ?        I<   17:26   0:00 [kworker/R-sync_wq]
root         6  0.0  0.0      0     0 ?        I<   17:26   0:00 [kworker/R-kvfree_rcu_reclaim]
root         7  0.0  0.0      0     0 ?        I<   17:26   0:00 [kworker/R-slub_flushwq]
root         8  0.0  0.0      0     0 ?        I<   17:26   0:00 [kworker/R-netns]
root         9  0.0  0.0      0     0 ?        I    17:26   0:00 [kworker/0:0-virtio_vsock]
197
parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$
```

Figure 2.1 – Process list (`ps aux`) and number of processes

```
top - 17:38:24 up 11 min, 1 user, load average: 0.09, 0.06, 0.01
Tasks: 196 total, 1 running, 195 sleeping, 0 stopped, 0 zombie
%Cpu(s): 3.7 us, 1.0 sy, 0.0 ni, 95.3 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3899.1 total, 1870.6 free, 1029.9 used, 1308.1 buff/cache
MiB Swap: 3899.0 total, 3899.0 free, 0.0 used. 2869.2 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2651	paralle+	20	0	3892544	282104	126600	S	5.0	7.1	0:27.26	gnome-shell
5295	paralle+	20	0	574380	56956	44016	S	1.7	1.4	0:02.06	gnome-terminal-
2807	paralle+	20	0	818908	26104	17784	S	0.7	0.7	0:04.95	prlcc
2820	paralle+	20	0	307992	5632	5120	S	0.7	0.1	0:02.01	prldnd
3258	paralle+	20	0	2822276	62592	46784	S	0.7	1.6	0:04.76	gjs
2428	paralle+	20	0	10836	5692	3772	S	0.3	0.1	0:01.02	dbus-daemon
2822	paralle+	20	0	153636	3296	3040	S	0.3	0.1	0:00.21	prlshprof
1	root	20	0	23052	13228	8620	S	0.0	0.3	0:00.55	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
9	root	20	0	0	0	0	I	0.0	0.0	0:00.38	kworker/0:0-virtio_vsock
11	root	20	0	0	0	0	I	0.0	0.0	0:00.29	kworker/0:1-events
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
16	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
17	root	20	0	0	0	0	S	0.0	0.0	0:00.08	ksoftirqd/0
18	root	20	0	0	0	0	I	0.0	0.0	0:00.28	rcu_preempt
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/0
20	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_gp_kthread_worker
21	root	rt	0	0	0	0	S	0.0	0.0	0:00.00	migration/0
22	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
23	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
24	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
25	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
26	root	rt	0	0	0	0	S	0.0	0.0	0:00.00	migration/1
27	root	20	0	0	0	0	S	0.0	0.0	0:00.07	ksoftirqd/1

Figure 2.2 – Live view of processes, CPU and memory in top

```
parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$ sleep 300 &

jobs
ps -ef | grep sleep
kill 5972
jobs
[2] 7093
[1] Done sleep 300
[2]+ Running sleep 300 &
paralle+ 7093 5302 0 17:46 pts/0 00:00:00 sleep 300
paralle+ 7095 5302 0 17:46 pts/0 00:00:00 grep --color=auto sleep
bash: kill: (5972) - No such process
[2]+ Running sleep 300 &
parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$
```

Figure 2.3 – Starting a background sleep process and finding its PID

```
parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$ kill 7093
jobs
[2]+ Running sleep 300 &
parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$ jobs
[2]+ Terminated sleep 300
parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$
```

Figure 2.4 – Killing the process and confirming it was terminated

```

parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$ sleep 300 &
PID=$!
ls /proc/$PID/
cat /proc/$PID/status | head -20
kill $PID
[2] 6146
attr          comm          fd          latency      mountinfo    oom_adj      root         smaps_rollup task
autogroup     coredump_filter fdinfo       limits       mounts       oom_score    sched        stack        timens_offsets
auxv          cpuset        gid_map      loginuid     mountstats   oom_score_adj schedstat    stat         timers
cgroup        cwd           io           map_files    net          pagemap      sessionid   statm        timerslack_ns
clear_refs    environ      ksm_merging_pages maps          ns           personality  setgroups   status       uid_map
cmdline       exe          ksm_stat     mem          numa_maps    projid_map   smaps       syscall      wchan
Name:         sleep
Umask:        0002
State:        S (sleeping)
Tgid:         6146
Ngid:         0
Pid:          6146
PPid:         5302
TracerPid:    0
Uid:          1000    1000    1000    1000
Gid:          1000    1000    1000    1000
FDSize:       256
Groups:       4 24 27 30 46 100 114 1000
NSTgid:       6146
NSpid:        6146
NSpgid:       6146
NSSid:        5302
Kthreadd:     0
VmPeak:       16532 kB
VmSize:       16532 kB
VmLck:        0 kB

```

Figure 2.5 – Process information exposed in /proc/<PID>/status

```

parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$ ps -p 1
  PID TTY          TIME CMD
   1 ?           00:00:00 systemd
[2]+  Terminated                  sleep 300

```

Figure 2.6 – Process with PID 1 is systemd

Interesting output:

```

$ cat /proc/$PID/status | head -20
Name:         sleep
State:        S (sleeping)
Pid:          6146
PPid:         5302
Uid:          1000    1000    1000    1000
VmPeak:       16532 kB
VmSize:       16532 kB
...

```

Starting and killing a background process:

```

$ sleep 300 &
[2] 7093
$ ps -ef | grep sleep
paralle+  7093    5302  0 17:46 pts/0    00:00:00 sleep 300
$ kill 7093
$ jobs
[2]+  Terminated                  sleep 300

```

(1) What is PID 1?

PID 1 is systemd: `ps aux` lists it as `/sbin/init splash`, and `ps -p 1` shows `systemd`. It is the first user-space process the kernel starts at boot. It launches all other services, and it adopts orphaned processes.

(2) How many processes on the idle VM?

About 196. `ps aux | wc -l` printed 197, which includes one header line, and `top` showed Tasks: 196 total, 1 running, 195 sleeping. Many of them are kernel threads, such as `kthreadd` and the `kworker/...` threads.

(3) What does State: say for the sleeping process?

State: S (sleeping). The process is waiting for its timer to expire and uses no CPU time until the kernel wakes it. Its PPid (5302) is my bash shell, which started it.

Part 3. Memory

```
parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$ free -h
cat /proc/meminfo | head -6
sleep 300 & PID=$!
grep VmRSS /proc/$PID/status
kill $PID
```

	total	used	free	shared	buff/cache	available
Mem:	3.8Gi	938Mi	1.9Gi	159Mi	1.3Gi	2.9Gi
Swap:	3.8Gi	0B	3.8Gi			
MemTotal:	3992628 kB					
MemFree:	2008044 kB					
MemAvailable:	3031940 kB					
Buffers:	43868 kB					
Cached:	1245204 kB					
SwapCached:	0 kB					
[2] 6356						
VmRSS:	1816 kB					

```
parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$
```

Figure 3.1 – RAM and swap usage (`free -h`, `/proc/meminfo`, `VmRSS`)

Interesting output:

\$ free -h						
	total	used	free	shared	buff/cache	available
Mem:	3.8Gi	938Mi	1.9Gi	159Mi	1.3Gi	2.9Gi
Swap:	3.8Gi	0B	3.8Gi			

(1) Total and free RAM:

Total is 3.8 GiB (MemTotal: 3992628 kB). Free is 1.9 GiB, and available is 2.9 GiB. "Available" is higher than "free" because the kernel uses about 1.3 GiB as disk cache, and it can reclaim that memory when programs need it.

(2) What is swap, and how much is configured?

Swap is disk space the OS uses as overflow for RAM. When memory runs low, it moves pages that are rarely used out to disk. This VM has 3.8 GiB of swap, and 0 B of it is used. It is the file `/swap.img` that appears in `ls -la /`, about 4 GB and readable only by root.

(3) VmRSS of sleep:

VmRSS: 1816 kB, about 1.8 MB. That surprised me, since the program does nothing. Most of it is the shared C library (libc) and the dynamic loader mapped into the process. Those pages are shared with other processes, so they cost almost no extra physical RAM. The virtual size (VmSize: 16532 kB) is even larger, because most of that address space is reserved but never actually loaded into RAM.

Part 4. Devices and storage

```
parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$ df -h
lsblk
du -sh ~/os-lab1
ls -l /dev | head
mount | head
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           390M  1.7M  389M   1% /run
/dev/sda2       62G   9.6G   49G  17% /
tmpfs           2.0G   0   2.0G   0% /dev/shm
tmpfs           5.0M   8.0K   5.0M   1% /run/lock
efivarfs        104K   4.8K  100K   5% /sys/firmware/efi/efivars
/dev/sda1       1.1G   6.4M   1.1G   1% /boot/efi
tmpfs           390M  156K  390M   1% /run/user/1000
[2]+  Terminated                  sleep 300
NAME MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0  7:0    0  68.9M  1 loop /snap/core22/2049
loop1  7:1    0    4K   1 loop /snap/bare/5
loop2  7:2    0  68.9M  1 loop /snap/core22/2086
loop3  7:3    0 230.5M  1 loop /snap/firefox/6563
loop4  7:4    0 231.5M  1 loop /snap/firefox/6696
loop5  7:5    0  91.7M  1 loop /snap/gtk-common-themes/1535
loop6  7:6    0   10M   1 loop /snap/snap-store/1271
loop7  7:7    0 493.5M  1 loop /snap/gnome-42-2204/201
loop8  7:8    0  42.9M  1 loop /snap/snapd/24787
loop9  7:9    0  44.2M  1 loop /snap/snapd/25205
loop10 7:10   0  552K   1 loop /snap/snapd-desktop-integration/316
sda     8:0    0   64G   disk
├─sda1   8:1    0    1G   0 part /boot/efi
└─sda2   8:2    0 62.9G   0 part /
sr0     11:0   1 1024M   0 rom
12K     /home/parallels/os-lab1
total 0
crw-r--r--  1 root    root    10, 235 Sep 28 17:26 autofs
drwxr-xr-x  2 root    root      360 Sep 28 17:26 block
drwxr-xr-x  2 root    root      80 Sep 28 17:26 bsg
crw-----  1 root    root    10, 234 Sep 28 17:26 btrfs-control
drwxr-xr-x  3 root    root      60 Jan  1 1970 bus
lrwxrwxrwx  1 root    root      3 Sep 28 17:26 cdrom -> sr0
drwxr-xr-x  2 root    root    3900 Sep 28 17:27 char
crw--w----  1 root    tty      5,  1 Sep 28 17:26 console
lrwxrwxrwx  1 root    root     11 Sep 28 17:26 core -> /proc/kcore
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
udev on /dev type devtmpfs (rw,nosuid,relatime,size=1938052k,nr_inodes=484513,node=755,inode64)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=000)
tmpfs on /run type tmpfs (rw,nosuid,nodev,noexec,relatime,size=399264k,node=755,inode64)
/dev/sda2 on / type ext4 (rw,relatime)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,inode64)
tmpfs on /run/lock type tmpfs (rw,nosuid,nodev,noexec,relatime,size=5120k,inode64)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate,memory_recursiveprot)
parallels@ubuntu-gnu-linux-24-04-3:~/os-lab1/demo$
```

Figure 4.1 – Filesystems, block devices, /dev entries and mounts

Interesting output:

```
$ lsblk
NAME MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda     8:0    0   64G   0 disk
├─sda1   8:1    0    1G   0 part /boot/efi
└─sda2   8:2    0 62.9G   0 part /
sr0     11:0   1 1024M   0 rom
```

(1) Which device is / mounted on?

/dev/sda2, an ext4 filesystem of 62 GB with 17% used (mount shows /dev/sda2 on / type ext4). The other partition, /dev/sda1, holds the EFI boot files at /boot/efi.

(2) One entry from /dev:

/dev/cdrom is a symbolic link to sr0, the VM's virtual CD/DVD drive. /dev/sda is the whole 64 GB virtual hard disk.

(3) "Everything is a file":

Disks (/dev/sda), the CD drive (/dev/sr0), the console, and even kernel memory (/dev/core -> /proc/kcore) all appear as files. That means ordinary tools like ls and cat can inspect hardware and the kernel.

Conclusions

The OS manages four resources: files, processes, memory and devices.

- `ls -l` let me see files, with their owners and permissions.
- `ps aux` let me see processes; about 196 were running.
- `free -h` let me see memory: 3.8 GiB of RAM and 3.8 GiB of swap.
- `lsblk` let me see devices: the 64 GB disk `sda`, split into `sda1` and `sda2`.